Incinerator discharge gases:
at least 4 air changes in the
chamber(s) and stack, but not
less than 15 seconds.
at least 4 air changes in the
chamber(s) and stack, but not
less than 15 seconds.
not less than 15 seconds after
the closing of the fuel oil
valve.
Minimum 6~ 02'

---CHUNK---

SOURCE: MEPC.59(33).pdf
chamber(s) and stack, but not
less than 15 seconds.
at least 4 air changes in the
chamber(s) and stack, but not
less than 15 seconds.
not less than 15 seconds after
the closing of the fuel oil
valve.
Minimum 6~ 02'
4.2 Outside surface of combustion chamber(s) should be shielded from contact such that people in normal work situations will not be exposed to extreme heat (20 0 e above ambient temperature) or direct contact of surface temperatures exceeding 60 o e. Examples for alternatives to accomplish this are a double jacket with an air flow in between or an expanded metal jacket.
W/9096D/ljt/EWP
RESOLUTION MEPC.59(33)
adopted on 30 October 1992
REVISED GUIDELINES FOR THE IMPLEMENTATION OF
ANNEX V OF MARPOL 73/78MEPC 33/20/Add.1
ANNEX 2
Page 10
4.3 Incinerating systems are to be operated with underpressure (negative pressure) in the combustion chamber such that no gases or smoke can leak out to the surrounding areas.
4.4 The incinerator should have warning plates attached in a prominent location on the unit, warning against unauthorized opening of doors to combustion chamber(s) during operation and against overloading the incinerator with garbage.

---CHUNK---

SOURCE: MEPC.59(33).pdf
4.4 The incinerator should have warning plates attached in a prominent location on the unit, warning against unauthorized opening of doors to combustion chamber(s) during operation and against overloading the incinerator with garbage.
4.5 The incinerator should have instruction plate(s) attached in a prominent location on the unit that clearly addresses the following:
4.5.1 Cleaning ashes and slag from the combustion chamber(s) and cleaning of

combustion air openings before starting the incinerator (where applicable).
4.5.2 Operating procedures and instructions. These should include proper start-up procedures, normal shut-down procedures, emergency shut-down procedures, and procedures for loading garbage (where applicable).
4.6 To avoid building up of dioxins, the flue gas should be shock-cooled to a maximum 350°C right after the incinerator.
5 Operating controls
5.1 The entire unit should be capable of being disconnected from all sources of electricity by means of one disconnect switch located near the incinerator. (See 3.14.2.1)
5.2 There should be an emergency stop switch located outside the compartment

---CHUNK---

SOURCE: MEPC.59(33).pdf
sources of electricity by means of one disconnect switch located near the incinerator. (See 3.14.2.1)
5.2 There should be an emergency stop switch located outside the compartment which stops all power to the equipment. The emergency stop switch should also be able to stop all power to the fuel pumps. If the incinerator is equipped with a flue gas fan, the fan should be capable of being restarted independently of the other equipment on the incinerator.
5.3 The control equipment should be so designed that any failure of the following equipment will prevent continued operations and cause the fuel supply to be cut off.
5.3.1 Safety thermostat/draft failure
5.3.1.1 A flue gas temperature controller, with a sensor placed in the flue gas duct, should be provided that will shut down the burner if the flue gas temperature exceeds the temperature set by the manufacturer for the specific design.
5.3.1.2 A combustion temperature controller, with a sensor placed in the combustion chamber, should be provided that will shut down the burner if the combustion chamber temperature exceeds the maximum temperature.

---CHUNK---

SOURCE: MEPC.59(33).pdf
design.
5.3.1.2 A combustion temperature controller, with a sensor placed in the combustion chamber, should be provided that will shut down the burner if the combustion chamber temperature exceeds the maximum temperature.
5.3.1.3 A negative pressure switch should be provided to monitor the draft and the negative pressure in the combustion chamber. The purpose of this negative pressure switch is to ensure that there is sufficient draft/negative pressure in the incinerator during operations. The circuit to the program relay for the burner will be opened and an alarm activated before the negative pressure rises to atmospheric pressure.
W/9096D/ljt/EWP
RESOLUTION MEPC.59(33)
adopted on 30 October 1992
REVISED GUIDELINES FOR THE IMPLEMENTATION OF
ANNEX V OF MARPOL 73/785.3.2 Flame failure/fuel oil pressure:
MEPC 33/20/Add.l
ANNEX 2

5.3.2.1 The incinerator should have a flame safeguard control consisting of a flame ~ensing element and assoc i ated equipment for shut down of the unit in the event of ignition failure a nd fl'~e 'fi:tilure'duringthe firing cycle. The flame safeguard control should be so designed that the failure of any

---CHUNK---

SOURCE: MEPC.59(33).pdf
flame ~ensing element and assoc i ated equipment for shut down of the unit in the event of ignition failure a nd fl'~e 'fi:tilure'duringthe firing cycle. The flame safeguard control should be so designed that the failure of any component will cause a safety shut down.
5.3.2.2 The flame safeguard control should be capable of closing the fuel valves in not more than 4 seconds after a flame failure.
5.3.2.3 The flame safeguard control should provide a trial-for-ignition period of not more that 10 seconds during which fuel may be supplied to establish flame. If flame is not established within 10 seconds, the fuel supply to the burners should be immediately shut off automatically.
5.3.2.4 Whenever the flame safeguard control has operated because of failure of ignition, flame failure, or failure of any component, only one automatic restart may be provided. If this is not successful then manual reset of the flame safeguard control should be required for restart.
5.3.2.5 Flame safeguard controls of the thermostatic type, such as stack switches and pyrostats operated by means of an open bimetallic helix, are prohibited.

---CHUNK---

SOURCE: MEPC.59(33).pdf
flame safeguard control should be required for restart.
5.3.2.5 Flame safeguard controls of the thermostatic type, such as stack switches and pyrostats operated by means of an open bimetallic helix, are prohibited.
5.3.2.6 If fuel oil pressure drops below that set by. the manufacturer, a failure and lock out of the program relay should result. This also applies to sludge oil used as a fuel. (Applies where pressure is important for the combustion process or a pump is not an integral part of the burner.)
5.3.3 Loss of power
If there is a loss of power to the incinerator control/alarm panel (not remote alarm panel), the system should shut down.
5.4 Fuel supply
Two fuel control solenoid valves should be provided in series in the fuel supply line to each burner. On multiple burner units, a valve on the main fuel supply line and a valve at each burner will satisfy this requirement. The valves should be connected electrically in parallel so that both operate simultaneously.
5.5 Alarms
5.5.1 An outlet for an audible alarm should be provided for connection to a local alarm system or a central alarm system. When a failure occurs, a

---CHUNK---

SOURCE: MEPC.59(33).pdf

simultaneously.
5.5 Alarms
5.5.1 An outlet for an audible alarm should be provided for connection to a local alarm system or a central alarm system. When a failure occurs, a visible indicator should show what caused the failure. (The indicator may cover more than one fault condition.)
5.5.2 The visible indicators should be designed so that, where failure is a safety related shutdown, manual reset is required.
W/ 9096DIlj tlEWP
RESOLUTION MEPC.59(33)
adopted on 30 October 1992
REVISED GUIDELINES FOR THE IMPLEMENTATION OF
ANNEX V OF MARPOL 73/78MEPC 33/20/Add.l
ANNEX 2
Page 12
5.6 After shutdown of the oil burner, prov~s~on should be made for the fire box to cool sufficiently. (As an example, of how this may be accomplished, the exhaust fan or ejector could be designed to continue to operate. This would not apply in the case of an emergency manual trip.)
6 Other requirements
6 . 1 Documentation
A complete instruction and maintenance manual with drawings, electric diagrams, spare parts list, etc., should be furnished with each incinerator.